

NO.5000 车载充电机 CAN 总线通讯规范 Ver1.2

CAN BUS COMMUNICATION SPECIFICATION

1.通讯规范 Communication Specification

数据链路层应遵循的原则

The principle for data link layer

总线通讯速率为：250Kbps

Communication speed for bus line: 250Kbps

数据链路层的规定主要参考 CAN2.0B 和 J1939 的相关规定

The provision for data link layer: Refer to the related regulation of CAN2.0B and J1939

使用 CAN 扩展帧的 29 位标识符并进行了重新定义，以下为 29 标识符的分配表：

Use and redefine 29 identifiers of CAN extended frame. The distribution of 29 identifiers are listed below

IDENTIFIER 11BITS											S R R E	I D E	IDENTIFIER EXTENSION 18BITS																	
PRIORI TY			R	D P	PDU FORMAT(PF)						S R R E	I D E	PF	PDU SPECIFIC(PS)								SOURCE ADDRESS(SA)								
3	2	1	1	1	8	7	6	5	4	3			2	1	8	7	6	5	4	3	2	1	8	7	6	5	4	3	2	1
2	2	2	2	2	2	2	2	2	1	1			1	1	1	1	1	1	1	1	9	8	7	6	5	4	3	2	1	0
8	7	6	5	4	3	2	1	0	9	8			7	6	5	4	3	2	1	0										

>优先级为 3 位，可以有 8 个优先级；R 一般固定为 0；DP 现固定为 0；8 位的 PF 为报

文的代码；8 位的 PS 为目标地址或组扩展；8 位的 SA 为发送此报文的源地址。

Priority has 3 bits so there can be 8 priorities. R is generally 0. DP is fixed at 0. 8-bit PF is the code for the message. 8-bit PS refers to destination address. 8-bit SA refers to the source address.

>接入网络的每一个节点都有名称和地址，名称用于识别节点的功能和进行地址仲裁，地址用于节点的数据通信

There is a name and an address for every node which accesses to the network. The name is used for nodes identification and address arbitration. The address is used for data communication to node.

>每个节点都至少有一种功能，可能会有多个节点具有相同的功能，也可能一个节点具有多个功能

Every node has at least one function. Multiple nodes might have the same function or one node might have multiple functions.

2.CAN 网络地址分配表 CAN Network Address Distribution

CAN 总线结点地址从 J1939 标准中定义的获得：

Obtain the node address of CAN Bus from the definition of J1939 Standard:

结点名称 Node Name	地址 SOURCE ADDRESS(SA)
电池管理系统 (BMS) Battery Management System	165 (0XA5)
逆变器控制系统 (ICS) Inverter Control System	182 (0XB6)
广播地址 (BCA) Broadcast Address	80 (0X50)

3.报文格式 Message Format

Message1: (ID:0X1806B6A5)

OUT	IN	ID				周期(ms) CYCLE TIME
BMS	ICS	P	R	DP	PF	500
		6	0	0	6	
数据						
位置	数据名					
BYTE0	设定逆变器输出电压高字节 Set Inverter Output Voltage High Byte		0.1V/bit 偏移量: 0			
BYTE1	设定逆变器输出电压低字节 Set Inverter Output Voltage Low Byte		例: Vset=2200, 对应电压为 220v 0.1V/byte offset:0 e.g. Vset=2200, its corresponding 220V			

BYTE2	保留 Reserved	保留 Reserved
BYTE3	保留 Reserved	保留 Reserved
BYTE4	控制 Control	0: 开启 Start 1: 关闭输出 Stop
BYTE5	保留 Reserved	
BYTE6	保留 Reserved	
BYTE7	保留 Reserved	

Message2: (ID:0X18FF50B6)

OUT	IN	ID				周期(ms) CYCLE TIME
ICS	BCA	P	R	DP	PF	500
		6	0	0	0XFF	
数据						
位置	数据名					
BYTE0	逆变器输出电压高字节 Output Voltage High Byte		0.1V/bit 偏移量: 0 例: Vout=2200, 对应电压为 220v 0.1V/byte offset:0 e.g. Vout=2200, its corresponding 220V			
BYTE1	逆变器输出电压低字节 Output Voltage Low Byte					
BYTE2	逆变器输出电流高字节 Output Current High Byte		0.1A/bit 偏移量: 0 例: Iset=280, 对应电压为 28.0A 0.1A/byte offset:0 e.g. Iout=582, its corresponding 58.2A			
BYTE3	逆变器输出电流低字节 Output Current Low Byte					
BYTE4	状态标志 1 Status Flags1					
BYTE5	状态标志 2 Status Flags2					
BYTE6	保留 Reserved		保留 Reserved			
BYTE7	系统温度 Temperature		散热器温度 单位 1℃ 偏移-40℃ e.g. BYTE7=120 Temp=80℃			

状态标志 1	标识	
Bit0	硬件故障 Hardware Failure	0: 正常。1: 硬件故障 0: Normal. 1: Hardware Failure
Bit1	温度 Temperature	0: 正常。1: 温度过高保护 0: Normal. 1: Over temperature protection
Bit2	输入电压 Input Voltage	0: 输入电压正常。1: 输入电压错误, 充电机停止工作 0: Input voltage is normal. 1. Input voltage is wrong, the charger will stop working.
Bit3	输入电流 Input current	0: 输入电压正常。1: 输入电压错误, 停止工作 0: Input current is normal. 1. Input current is wrong, the charger will stop working.
Bit4	输出电压 Out Voltage	0: 输出电压正常。1: 输出电压错误, 停止工作 0: Out voltage is normal. 1.Out voltage is wrong, the charger will stop working.
Bit5	输出电流 Out current	0: 输出电流正常。1: 输出电流错误, 停止工作 0: Out current is normal. 1.Out current is wrong, the charger will stop working.
Bit6	启动状态 Starting state	0: 逆变器进入启动状态。 0: Starts charging 1: 处于关闭状态。 1: Charger stays turned off
Bit7	通信状态 Communication State	0: 通信正常 0: Communication is normal 1: 通信接收超时 1: Communication receive time-out

状态标志 2	标识	
Bit0	母线电压 Hardware Failure	0: 母线电压正常。1: 母线电压错误, 停止工作 0: Bus voltage is normal. 1: Bus voltage is wrong, the charger will stop working.
Bit1	母线电流 Temperature	0: 输出电流正常。1: 输出电流错误, 停止工作 0: Bus current is normal. 1: Bus current is wrong, the charger will stop working.
Bit3	保留 Reserved	保留 Reserved
Bit4	保留 Reserved	保留 Reserved
Bit5	保留 Reserved	保留 Reserved
Bit6	保留 Reserved	保留 Reserved
Bit7	保留 Reserved	保留 Reserved

Message3: (ID:0X18FE50B6)

OUT	IN	ID				周期(ms) CYCLE TIME
ICS	BCA	P	R	DP	PF	500
		6	0	0	FE	
数据						
位置	数据名					
BYTE0	逆变器输入电压高字节 Inverter Input Voltage High Byte		0.1V/bit 偏移量: 0 例: Vin=3000, 对应电压为 300v 0.1V/byte offset:0 e.g. Vin=3000, its corresponding 300V			
BYTE1	逆变器输入电压低字节 Inverter Input Voltage Low Byte					
BYTE2	逆变器输入电流高字节 Inverter Input Current High Byte		0.1A/bit 偏移量: 0 例: Iin=280, 对应电压为 28.0A 0.1A/byte offset:0 e.g. Iin=280, its corresponding 28A			
BYTE3	逆变器输入电流低字节 Inverter Input Current Low Byte					
BYTE4	保留 Reserved					
BYTE5	保留 Reserved					
BYTE6	保留 Reserved					
BYTE7	保留 Reserved					

4.工作方式 Workflow

>BMS 固定间隔时间 500ms 发送控制信息（报文 1）到充电机，逆变器接收到信息以后根据报文数据的电压电流设置来工作。如果 6 秒接收不到报文，则进入通信错误状态，关闭输出。

The BMS sends operating information (Message 1) to interval at fixed interval of 500ms. After receiving the message, the interval will work under the Voltage and Current in Message. If the Message is not received within 5s, it will enter into communication error state and stop charging.

>充电机每隔 500ms 发送广播信息（报文 2），显示仪表可以根据信息显示逆变器状态。

The inverter send broadcast message (Message 2) at intervals of 500ms. The display meter can show the status of the interval according to up-to-date information.

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Add error .VERSION 1.1 2024/9/23

Add input voltage and input current.VERSION 1.2 2025/10/23